

AIOps Module 1 Assignment: Executive Summary

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Question 1: Technical Debt Diagnosis

- **Hidden technical debt categories:**
 - **Entanglement (CACE):** Delivery-time and recommendation features are tightly coupled, so tweaking one silently broke the other, even though they seem unrelated.
 - **Undeclared Consumers:** Marketing dashboard is quietly reading the model's output table with no contract, so the ML team can't change the output without maybe breaking their dashboard.
 - **Pipeline Jungles:** Training relies on an unmanageable, unorchestrated chain of undocumented shell scripts.
- **Mitigation:**
 - For (c), I would use a workflow orchestration tool like **Apache Airflow** to untangle the pipeline jungle. Rather than blindly chaining 14 shell scripts, Airflow allows the ML team to define the training workflow as a Directed Acyclic Graph (DAG). This provides explicit task dependencies, automated retries for failed steps, and proper logging, transforming a brittle sequence of scripts into a maintainable, observable pipeline.

Question 2: Applied MLflow Tracking

- **Best Configuration:** The optimal run (`mlp-1r0.001-bs32`) achieved a peak validation accuracy of 94.1% using a learning rate of 0.001 and a batch size of 32.
- **Overfitting Evidence:** The model exhibited severe overfitting. During the optimal run, training loss plummeted to a near-perfect 0.002, yet validation accuracy permanently plateaued. This massive divergence indicates the network memorized the training data rather than generalizing to unseen data.
- **Hyperparameter Impact:** Learning rate was the dominant driver of performance, causing a ~4% variance in accuracy, whereas toggling batch size only shifted outcomes by 1-2%. (*Note: The MLflow logging script and comparison table are located in the Q2 repository folder.*)

Question 3: Data Versioning (DVC)

- Successfully initialized DVC to track the official image dataset (Version 1).
- Simulated dataset drift by removing targeted files (Version 2) and executed a verifiable `dvc checkout` rollback to successfully restore the exact Version 1 state.

Question 4: Reproducibility & Collaboration

- **Partner A** established an isolated `environment.yml` and base MLflow tracking script in a newly initialized, shared repository.
- **Partner B** successfully cloned the repository, replicated the exact environment, and logged an independent MLflow run to guarantee cross-machine reproducibility.